Motor and Hall

File: MotorHall.kicad_sch

Matrix

File: Matrix.kicad_sch

MCU

File: MCU.kicad_sch

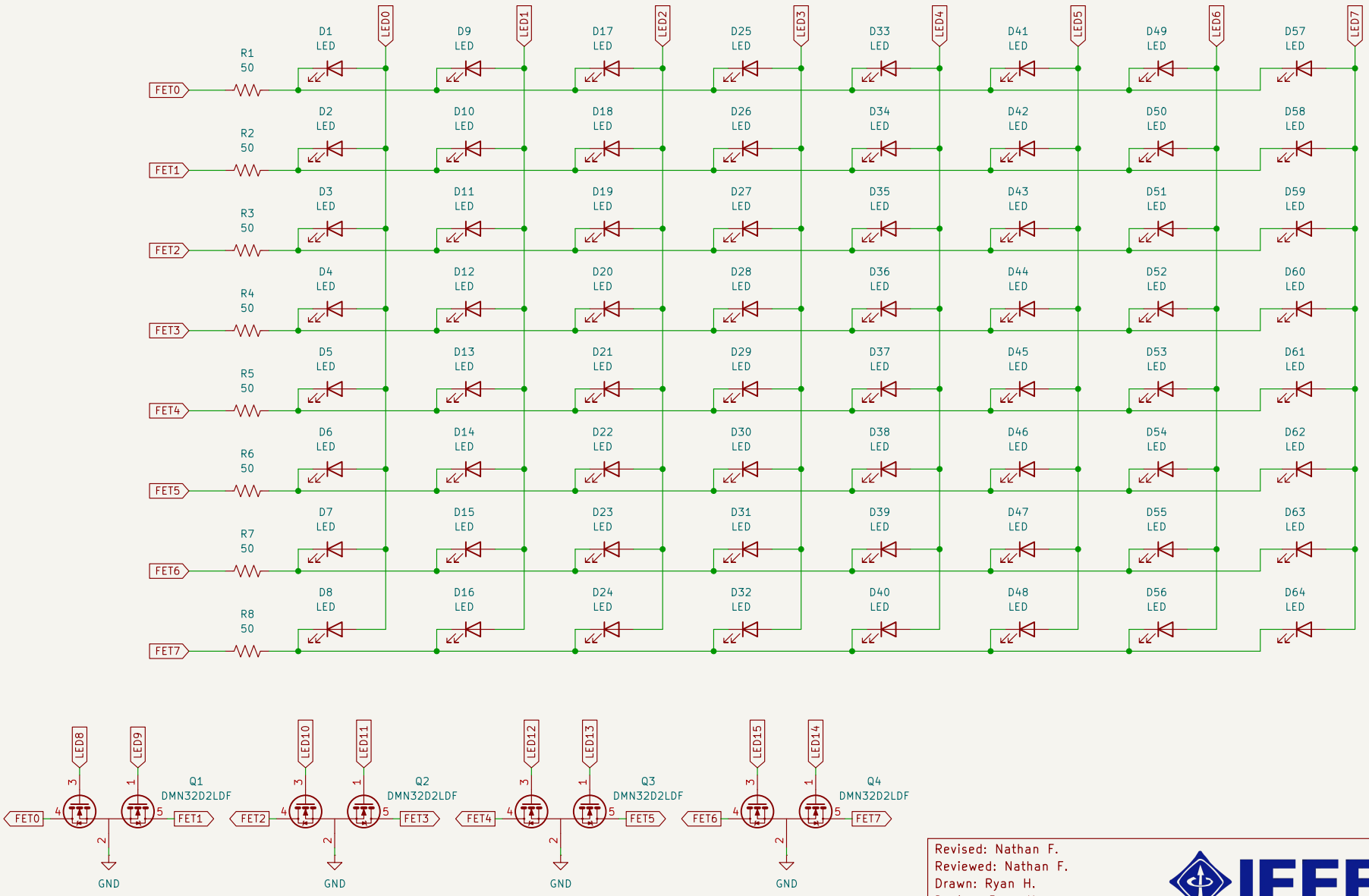
Reviewed: Nathan F.
Drawn: Ryan H.
Design: Ryan H.
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Sheet: /
File: VolumetricDisplay.kicad_sch

Title: Volumetric Display

Size: A4	Date: 2026-08-11	**Rev: X0**
KiCad E.D.A. 10.0.1		Id: 1/4



10 mA per LED:
3V3 drive, 2V drop
1.3V / 10 mA = 130 Ohms

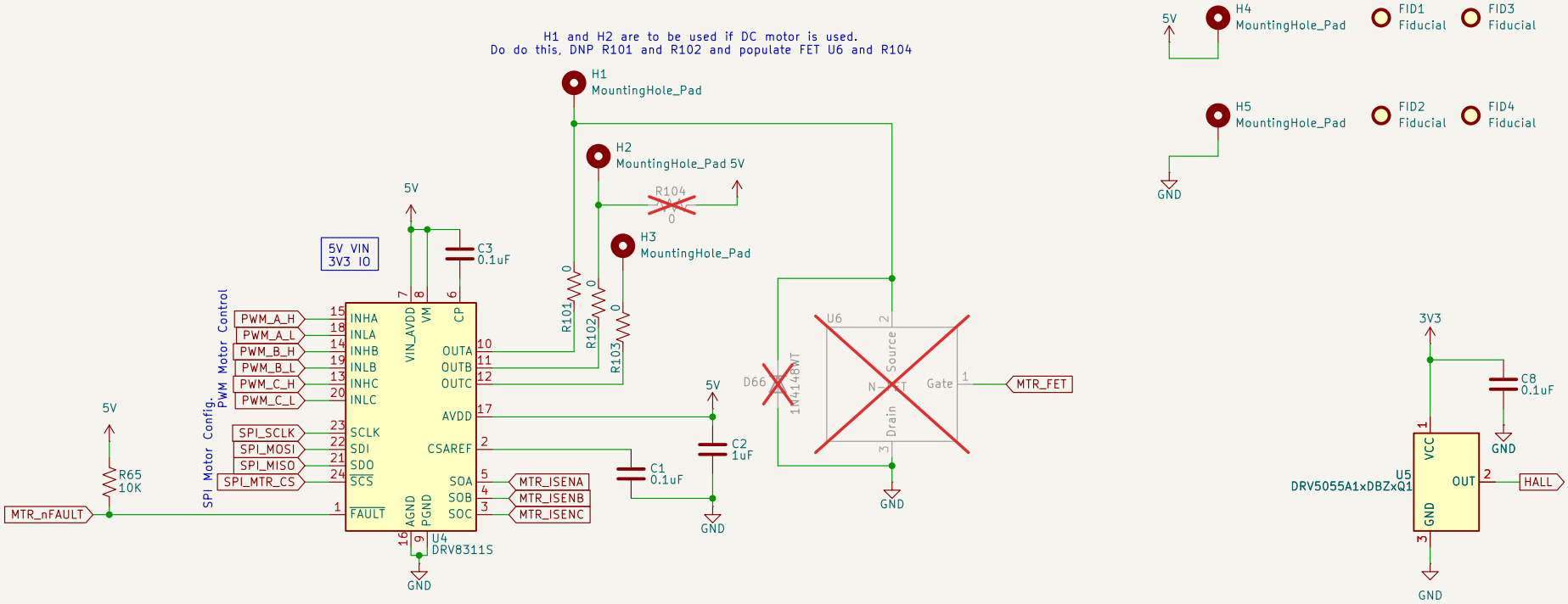
Revised: Nathan F.
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Title: Volumetric Display

Size: A4	Date:	**Rev: X0**
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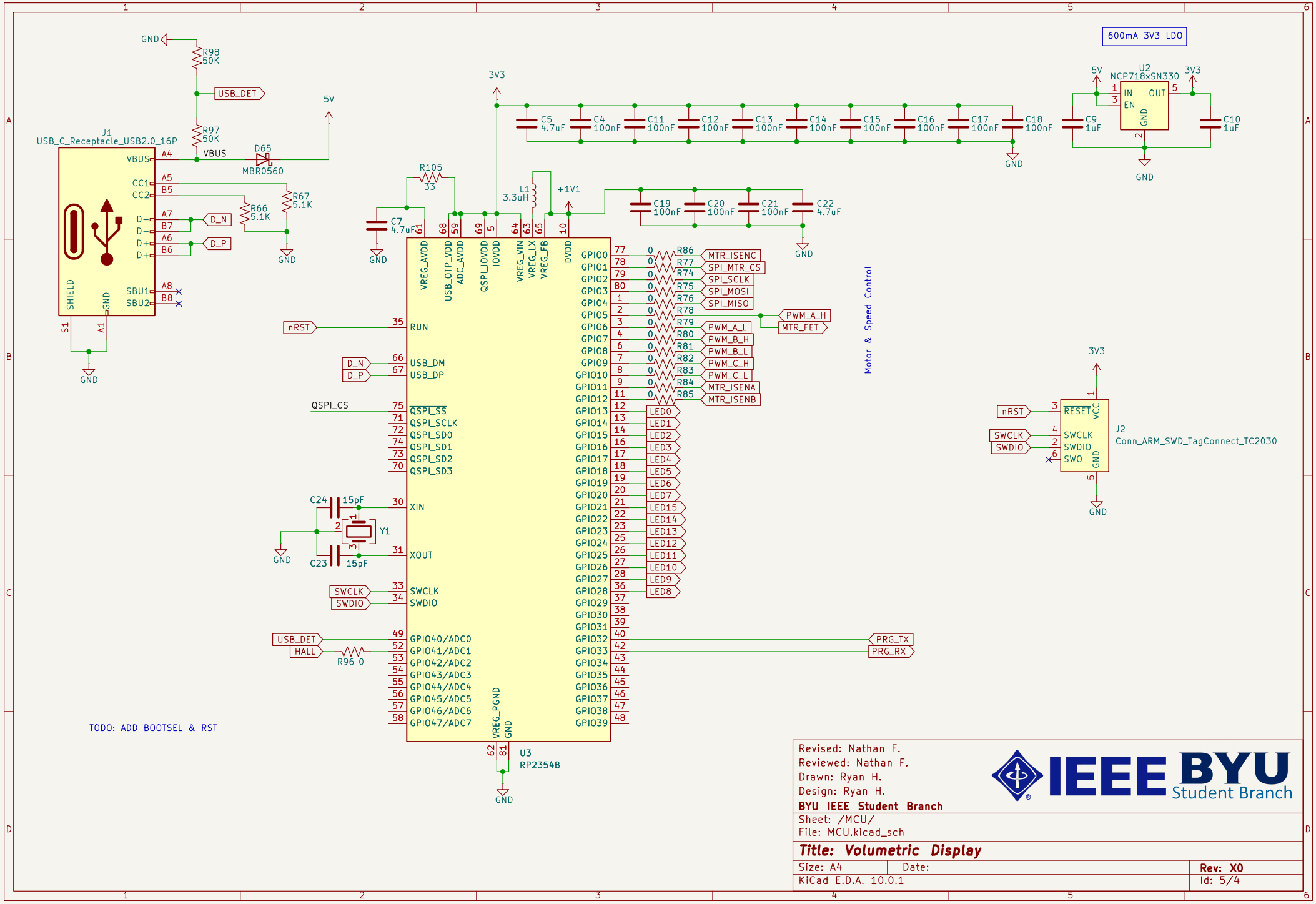
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Sheet: /Motor and Hall/
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Size: A4	Date:	**Rev: X0**
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600mA 3V3 LDO
U2
NCP718xSN330
5V
3V3
IN
OUT
EN
GND
C9 1uF
C10 1uF
J1
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D+
SHIELD
SBU1
SBU2
R98 50K
R97 50K
USB_DET
D65
MBR0560
R66 5.1K
R67 5.1K
D_N
D_P
S1
A1
C5 4.7uF
C4 100nF
C11 100nF
C12 100nF
C13 100nF
C14 100nF
C15 100nF
C16 100nF
C17 100nF
C18 100nF
R105 33
C7 4.7uF
L1 3.3uH
+1V1
C19 100nF
C20 100nF
C21 100nF
C22 4.7uF
VREG_AVDD
USB_OTP_VDD
ADC_AVDD
QSPI_IOVDD
IOVDD
VREG_VIN
VREG_LX
VREG_FB
DVDD
RUN
nRST
USB_DM
USB_DP
QSPI_SS
QSPI_SCLK
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_CS
XIN
XOUT
C24 15pF
C23 15pF
Y1
SWCLK
SWDIO
GPIO40/ADC0
GPIO41/ADC1
GPIO42/ADC2
GPIO43/ADC3
GPIO44/ADC4
GPIO45/ADC5
GPIO46/ADC6
GPIO47/ADC7
HALL
R96 0
VREG_PGND
U3
RP2354B
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
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GPIO28
GPIO29
GPIO30
GPIO31
GPIO32
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
GPIO39
R86 MTR_ISENC
R77 SPI_MTR_CS
R74 SPI_SCLK
R75 SPI_MOSI
R76 SPI_MISO
R78
R79 PWM_A_L
R80 PWM_B_H
R81 PWM_B_L
R82 PWM_C_H
R83 PWM_C_L
R84 MTR_ISENA
R85 MTR_ISENB
PWM_A_H
MTR_FET
LED0
LED1
LED2
LED3
LED4
LED5
LED6
LED7
LED15
LED14
LED13
LED12
LED11
LED10
LED9
LED8
PRG_TX
PRG_RX
Motor & Speed Control
J2
Conn_ARM_SWD_TagConnect_TC2030
RESET
VCC
SWCLK
SWDIO
SWO
TODO: ADD BOOTSEL & RST
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